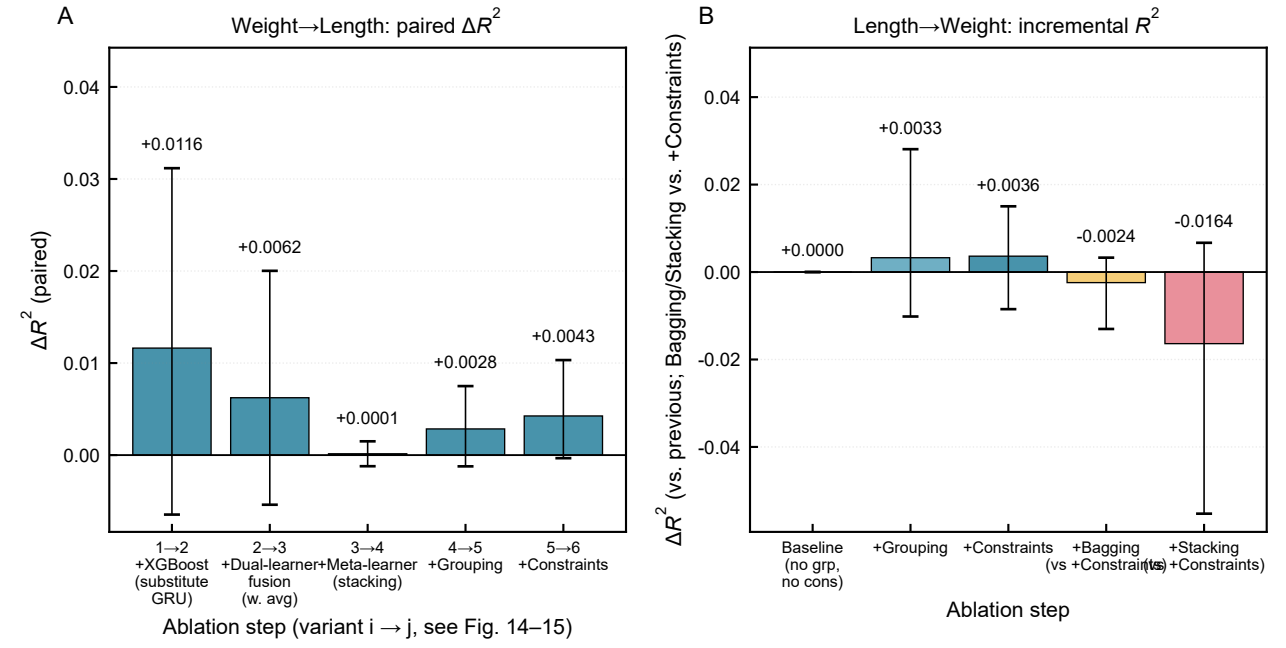
A
Weight→Length: paired ΔR²
ΔR² (paired)
0.04
0.03
0.02
0.01
0.00
+0.0116
+0.0062
+0.0001
+0.0028
+0.0043
1→2 +XGBoost (substitute GRU)
2→3 +Dual-learner fusion (w. avg)
3→4 +Meta-learner (stacking)
4→5 +Grouping
5→6 +Constraints
Ablation step (variant i → j, see Fig. 14–15)
B
Length→Weight: incremental R²
ΔR² (vs. previous; Bagging/Stacking vs. +Constraints)
0.04
0.02
0.00
-0.02
-0.04
+0.0000
+0.0033
+0.0036
-0.0024
-0.0164
Baseline (no grp, no cons)
+Grouping
+Constraints
+Bagging (vs +Constraints)
+Stacking (vs +Constraints)
Ablation step